SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

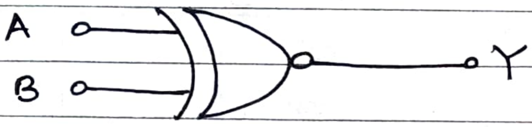
DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – IV – Boolean Algebra

(h) Exclusive NOR (EX-NOR) gate

The output is high only when even number of ones at the input or all inputs are high



$Y = \overline{A \oplus B}$

$= \bar{A}\cdot\bar{B} + A\cdot B$

Input A	Input B	Output Y
0	0	1
0	1	0
1	0	0
1	1	1

Application:

It is used to implement even parity generator, Comparator, even parity Checker etc.

2. Boolean Algebra

Video Link - https://youtu.be/C86U57EWwBc

Boolean algebra is a mathematical System that defines a series of logical operations (AND, OR, NOT) performed on sets of Variables (a, b, c, ...). When Stated in this form, the expression is called a Boolean equation or Switching equation

(a) Boolean Algebra Terminology

(i) Variable : The Symbol which represent an arbitary elements of an Boolean algebra is known as Variable.

(ii) Constant : In expression $Y = A + 1$, A is Variable and 1 is Constant. The Constant term may be 0 or 1

(iii) Complement : A Complement of a variable is represented by a bar over the letter

If $A = 1$, $\bar{A} = 0$

(iv) Literal : Each occurence of Variable in Boolean function either in Complemented or an uncomplemented form is called literal

(v) Boolean Function : Boolean expressions are Constructed by Connecting the Boolean Constants and Variables with the Boolean operations. These Boolean expressions are also known as Boolean formulae.

3. Axiomatic Definitions of Boolean Algebra

The postulates of a mathematical System form the basic assumption from which it is possible to deduce the theorems, laws and properties of the System. Boolean algebra is formulated by a defined set of elements, together with two binary operators, + and ·, provided that the given postulates are Satisfied.

(a) Closure Property

Video Link - https://youtu.be/B78GwEBQEeU

Closure (a) : Closure with respect to the operator + :
When two binary elements are operated by operator + the result is a Unique binary element

Closure (b) : Closure with respect to the operator · (dot) :
when two binary elements are operated by operator · (dot), the result is a unique binary element

Example :

$A, B \in S$

$C = A + B$, $C \in S$ is true

(b) Identity Property

$1 \cdot A = A \cdot 1 = A$

$I * x = x * I = x$

$I + x = x + I = x$

$I = 1$ for $*$

$I = 0$ for $+$

(c) Commutative Property

$A + B = B + A$

$A \cdot B = B \cdot A$

(d) Distributive Property

$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

$A + (B \cdot C) = (A + B) \cdot (A + C)$

(e) Associative Property

$$A + (B + C) = (A + B) + C$$
$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$

(f) Complement Property

$$A = 1, \quad \bar{A} = 0$$
$$A \cdot \bar{A} = 0$$
$$A + \bar{A} = 1$$

(g) Idempotency Property

$$A + A = A$$
$$A \cdot A = A$$

(h) Absorption Property

(i) $A + AB = A(1 + B)$

$A + AB = A \quad [\because 1 + B = 1]$

(ii) $A(A + B) = AA + AB$

$= A + AB$

$A(A + B) = \cancel{AB}\ A$

(i) Involution Property

Double inversion of any variable results in the original variable

$$\bar{\bar{A}} = A$$

4. Boolean Theorems

Video Link - https://youtu.be/ESLkJt1W51E

1. Single Variable Theorems
2. Multi Variable Theorems

(i) Single Variable Theorem

These include AND, OR, and NOT operations with a single input variable

(a) Law of Intersection – AND Laws

$x \cdot 0 = 0$	$x=1$	$1 \cdot 0$	$= 0$
	$x=0$	$0 \cdot 0$	$= 0$
$x \cdot 1 = x$	$x=1$	$1 \cdot 1$	$= 1$
	$x=0$	$0 \cdot 1$	$= 0$
$x \cdot x = x$	$x=1$	$1 \cdot 1$	$= 1$
	$x=0$	$0 \cdot 0$	$= 0$
$x \cdot \bar{x} = 0$	$x=0$	$0 \cdot 1$	$= 0$
	$x=1$	$1 \cdot 0$	$= 0$

(b) Law of Union - OR Laws

$x + 0 = x$

$x + 1 = 1$

$x + x = x$

$x + \bar{x} = 1$

(ii) Multivariable Theorems

These are theorems involving more than one variable

(a) Commutative Laws

$$x + y = y + x$$

$$x \cdot y = y \cdot x$$

Proof:

x	y	$x+y$	xy	$y \cdot x$	$y+x$
0	0	0	0	0	0
0	1	1	0	0	1
1	0	1	1	0	1
1	1	1	1	1	1

(b) Associative Law

$$x + (y+z) = (x+y) + z$$

$$x \cdot (yz) = (xy) \cdot z$$

(c) Distributive Law

$$x(y+z) = xy + xz$$

$$x + (yz) = (x+y) \cdot (x+z)$$

Proof

x y z	xy	xz	y+z	x(y+z)	xy + xz
0 0 0	0	0	0	0	0
0 0 1	0	0	1	0	0
0 1 0	0	0	1	0	0
0 1 1	0	0	1	0	0
1 0 0	0	0	0	0	0
1 0 1	0	1	1	1	1
1 1 0	1	0	1	1	1
1 1 1	1	0	1	1	1

$$x + yz = (x+y)(x+z)$$

Proof

$$x + yz = (x+y) \cdot (x+z)$$

$$= xx + xz + yx + yz$$

$$= x(1 + z + y) + yz$$

$$= x + yz \qquad [\because y+1 = 1$$

$$1+z = 1]$$

(d) Law of Absorption

(i) $x(x+y)\ x$

Proof:

$$x(x+y) = xx + x \cdot y$$

$$= x(1+y)$$

$$= x \qquad [\because 1+y = 1]$$

(ii) $x + xy = x$

Proof:

$$x + xy = x(1+y)$$

$$= x \qquad [\because 1+y = 1]$$

(iii) $x(\bar{x}+y) = x \cdot y$

Proof:

$$x(\bar{x}+y) = x\bar{x} + xy$$
$$= 0 + xy$$
$$= xy \qquad [\because x\bar{x} = 0]$$

(iv) $xy + \bar{y} = x + \bar{y}$

Proof:

$$xy + \bar{y} = xy + \bar{y}(x+1)$$
$$= xy + x\bar{y} + \bar{y}$$
$$= x(y+\bar{y}) + \bar{y}$$
$$= x(1) + \bar{y}$$
$$= x + \bar{y}$$

(v) $x\bar{y} + y = x + y$

Proof:

$$x\bar{y} + y = x\bar{y} + y(x+1)$$
$$= x\bar{y} + xy + y$$
$$= x(\bar{y}+y) + y$$
$$= x(1) + y$$
$$= x + y$$

(e) Consensus Law

(i) Theorem 1:

$$xy + \bar{x}z + yz = xy + \bar{x}z$$

Proof:

$$xy + \bar{x}z + yz = xy + \bar{x}z + yz(x+\bar{x})$$
$$= xy + \bar{x}z + xyz + \bar{x}yz$$
$$= xy(1+z) + \bar{x}z(1+y)$$
$$= xy(1) + \bar{x}z(1)$$
$$= xy + \bar{x}z$$

(ii) Theorem 2 :

$$(x+y)(\bar{x}+z)(y+z) = (x+y)(\bar{x}+z)$$

Proof :

$$(x+y)(\bar{x}+z)(y+z) = (x\bar{x} + xz + y\bar{x} + yz)(y+z)$$

$$= x\bar{x}y + x\bar{x}z + xzy + xz\cdot z + y\bar{x}\cdot y + y\bar{x}\cdot z + yyz + yzz$$

$$= 0 + 0 + xyz + xz + \bar{x}y + \bar{x}yz + yz + yz$$

$$= \bar{x}y + yz(x+\bar{x}) + yz + xz$$

$$= \bar{x}y + yz(1) + yz + xz$$

$$= \bar{x}y\cancel{1} + yz(1+yz) + xz$$

$$= \bar{x}y + yz + xz$$

$$= (\bar{x}+z)(x+y)$$

(f) DeMorgan's Theorem

Video Link - https://youtu.be/nxZGBQ5sleQ

DeMorgan's Suggested two theorems that form an important part of Boolean algebra.

In the equation form,

1. $\overline{AB} = \bar{A} + \bar{B}$
2. $\overline{A+B} = \bar{A}\cdot\bar{B}$

(i) $\overline{AB} = \bar{A} + \bar{B}$

The complement of a product is equal to the sum of the complements

A	B	$\overline{AB}$	$\bar{A}$	$\bar{B}$	$\bar{A}+\bar{B}$
0	0	1	1	1	1
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	0	0	0

(ii) $\overline{A+B} = \bar{A} \cdot \bar{B}$

The complement of a sum is equal to the product of the complements

A	B	$\bar{A}$	$\bar{B}$	$\overline{A+B}$	$\bar{A}\,\bar{B}$
0	0	1	1	1	1
0	1	1	0	0	0
1	0	0	1	0	0
1	1	0	0	0	0

5. Principle of Duality

The principle of duality theorem states that, Starting with a Boolean relation, you can derive another Boolean relation by

→ Changing each OR sign to an AND sign
→ Changing each AND sign to an OR sign
→ Complementing any 0 or 1 to ~~an OR~~ appearing in the expression

Example:

Dual relation $A + \bar{A} = 1$ is $\bar{A} \cdot A = 0$

Video Link - https://youtu.be/uUAed9_izWY

Postulates and Basic Theorems of Boolean Algebra

Postulates	(a)	(b)
Postulates 1	Result of each operation is 0 or 1 $\in S$	
2 (Identity)	$A + 0 = A$	$A \cdot 1 = A$
3 (Commutative)	$A + B = B + A$	$A \cdot B = B \cdot A$
4 (Distributive)	$A(B+C) = AB + BC$	$A + BC = (A+B)(A+C)$
5 (Complement)	$A + \bar{A} = 1$	$A \cdot \bar{A} = 0$
Theorems		
1 - Idempotency	$A + A = A$	$A \cdot A = A$
2 -	$A + 1 = 1$	$A \cdot 0 = 0$
3 - Involution	$\bar{\bar{A}} = A$	
4 - Absorption	$A + AB = A$	$A(A+B) = A$
5 -	$A + \bar{A}B = A + B$	$A \cdot (\bar{A} + B) = AB$
6 - Associative	$A + (B+C) = (A+B)+C$	$A(BC) = (AB)C$

6. Reduction of Switching equation by Boolean Algebra

Solved Problem #01

Video Link - https://youtu.be/bi8M1PONRlc

Prove that $A + \bar{A}B = A + B$

Solution :

$A + \bar{A}B = A + AB + \bar{A}B$ [$\because$ Absorption property $A + AB = A$]

$= A + B(A + \bar{A})$ [$\because$ Complement property $A + \bar{A} = 1$]

$= A + B \cdot 1$ [$\because$ $B \cdot 1 = B$ Identity property]

$= A + B$

$\therefore A + \bar{A}B = A + B$ proved

Solved Problem #02

Prove that $A \cdot (\bar{A} + B) = AB$

Solution:

$A \cdot (\bar{A} + B) = (A + AB) \cdot (\bar{A} + B)$ [$\because$ Absorption property $A + AB = A$]

$= A\bar{A} + AB + A\bar{A}B + ABB$

$= 0 + AB + A\bar{A}B + ABB$

$= 0 + AB + 0 + ABB$ [$\because$ Complement property $A \cdot \bar{A} = 0$]

$= AB + ABB$

$= AB + AB$

$= AB$

$\therefore A(\bar{A} + B) = AB$ proved

Solved Problem #03

Video Link - https://youtu.be/7K7ei_yPMX4

Apply De Morgan's theorem to simplify $\overline{A + B\bar{C}}$

Solution :

$\overline{A + B\bar{C}} = \bar{A} \cdot \overline{B\bar{C}}$

$= \bar{A} \cdot (\bar{B} + C)$

$= \bar{A}\bar{B} + \bar{A} \cdot C$

Solved Problems #04 Video Link - https://youtu.be/0FOk6QAXIF8

Simplify $\bar{A}BC\bar{D} + BC\bar{D} + B\bar{C}\bar{D} + B\bar{C}D$

Solution:

$$\bar{A}BC\bar{D} + BC\bar{D} + B\bar{C}\bar{D} + B\bar{C}D$$

$$= BC\bar{D}(A+1) + B\bar{C}\bar{D} + B\bar{C}D$$

$$= BC\bar{D}\cdot 1 + B\bar{C}\bar{D} + B\bar{C}D \qquad [\because A+1=1]$$

$$= B\bar{D}(C+\bar{C}) + B\bar{C}D$$

$$= B\bar{D}(1) + B\bar{C}D \qquad [\because C+\bar{C}=1]$$

$$= B\bar{D} + B\bar{C}D$$

$$= B(\bar{D} + \bar{C}D)$$

$$= B(\bar{D}+\bar{C}) \qquad [\because A+\bar{A}B = A+B,\ \bar{D}+D\bar{C} = \bar{D}+\bar{C}]$$

$\therefore$ Answer is $B(\bar{D}+\bar{C})$

Solved Problem #05 Video Link - https://youtu.be/WMJ7oCnwqac

Simplify $AB + \overline{AC} + A\bar{B}C(AB+C)$

Solution

$$AB + A\bar{C} + A\bar{B}C(AB+C) =$$

$$= AB + \overline{AC} + AAB\bar{B}C + A\bar{B}CC$$

$$= AB + \overline{AC} + 0 + A\bar{B}C \qquad [\because B\cdot\bar{B} = 0,\ C\cdot C = C]$$

$$= AB + \bar{A} + \bar{C} + A\bar{B}C \qquad [\because \text{De Morgan's Theorem } \overline{AB} = \bar{A}+\bar{B}]$$

$$= \bar{A} + B + \bar{C} + A\bar{B}C \qquad [\because A+\bar{A}B = A+B]$$

$$= \bar{A} + A\bar{B}C + B + \bar{C} \qquad [\bar{A} + A\bar{B}C = \bar{A}+\bar{B}C]$$

$$= \bar{A} + \bar{B}C + B + \bar{C} \qquad [\because A+\bar{A}B = A+B]$$

$$= \bar{A} + B + \bar{C} + \bar{B}C$$

$$= \bar{A} + B + \bar{C} + \bar{B} \qquad [\because A+\bar{A}B = A+B,\ \bar{C}+C\bar{B} = \bar{C}+\bar{B}]$$

$$= \bar{A} + \bar{C} + B + \bar{B}$$

$$= \bar{A} + \bar{C} + 1 \qquad [\because B+\bar{B} = 1]$$

$$= \bar{A} + \bar{C} + 1$$

$$= 1 \qquad [\because A+1 = 1]$$

Answer is 1

Solved Problem #06

Simplify $\bar{A}B + \bar{A}B\bar{C} + \bar{A}BCD + \bar{A}B\overline{CD}E$

Solution:

$$\bar{A}B + \bar{A}B\bar{C} + ABCD + AB\overline{CD}E = \bar{A}B(1 + \bar{C} + CD + \overline{CD}E)$$

$$= \bar{A}B(1) \quad [\because A+1=1]$$

$$= \bar{A}B$$

Solved Problem #07

Video Link - https://youtu.be/Yrppa-Zzxv0

Simplify $(P+Q+R)(\bar{P}+\bar{Q}+\bar{R})P$

Solution:

$$(P+Q+R)(\bar{P}+\bar{Q}+\bar{R})P$$

$$= (P\bar{P} + P\bar{Q} + P\bar{R} + Q\bar{P} + Q\bar{Q} + Q\bar{R} + R\bar{P} + R\bar{Q} + R\bar{R})P$$

$$= (0 + P\bar{Q} + P\bar{R} + Q\bar{P} + 0 + Q\bar{R} + R\bar{P} + R\bar{Q} + 0)P$$

$$[\because A \cdot \bar{A} = 0]$$

$$= PP\bar{Q} + PP\bar{R} + PQ\bar{P} + PQ\bar{R} + PR\bar{P} + PR\bar{Q}$$

$$= P\bar{Q} + P\bar{R} + PQ\bar{R} + PR\bar{Q}$$

$$= P\bar{Q} + P\bar{R} + PQ\bar{R} + P\bar{Q}R$$

$$= P\bar{Q}(1+P) + P\bar{R}(1+Q)$$

$$= P\bar{Q}(1) + P\bar{R}(1) \quad [\because 1+A=1]$$

$$= P\bar{Q} + P\bar{R}$$

Answer is $P\bar{Q} + P\bar{R}$

Solved Problem #08

Simplify $xyz + \bar{x}y + xy\bar{z}$

Solution:

$$xyz + \bar{x}y + xy\bar{z} = xy(z+\bar{z}) + \bar{x}y$$

$$= xy(1) + \bar{x}y \qquad [\because A+\bar{A} = 1]$$

$$= y(x+\bar{x})$$

$$= y(1) \qquad [\because A+\bar{A} = 1]$$

$$= y$$

Answer is y

Solved Problem #09

Simplify $\overline{(A+B)}\;\overline{(\bar{A}+B)}$

Solution:

$$\overline{(A+B)}+\overline{(\bar{A}+B)} = \bar{A}\cdot\bar{B} + A\cdot\bar{B}$$

$$= \bar{B}(\bar{A}+A)$$

$$= \bar{B}(1) \qquad [\because A+\bar{A} = 1]$$

$$= \bar{B}$$

Solved Problem #10

$$\overline{((\overline{A\bar{B}+ABC}) + A(B+A\bar{B}))}$$

Solution:

$$\overline{((\overline{A\bar{B}+ABC}) + A(B+A\bar{B}))}$$

$$= (\overline{\overline{A\bar{B}+ABC}})(\overline{A(B+A\bar{B})})$$

$$= (A\bar{B}+ABC)(\overline{A(B+A)}) \qquad [\because A+\bar{A}B = A+B]$$

$$= (A\bar{B}+ABC)(\bar{A}+(\overline{B+A}))$$

$$= (A\bar{B}+ABC)(\bar{A}+\bar{B}\cdot\bar{A})$$

$$= A(\bar{B}+BC)\,\bar{A}(1+\bar{B})$$

$$= A(\bar{B}+C)\cdot\bar{A} \qquad [A+\bar{A}B = A+B \;\&\; 1+\bar{B} = 1]$$

$$= 0 \qquad [\because A\cdot\bar{A} = 0]$$

Solved Problem #11

Simplify $A\bar{B}C + \bar{A}BC + ABC$

Solution :

$$A\bar{B}C + \bar{A}BC + ABC \cancel{= A\bar{B}C + AB(}$$

$$= A\bar{B}C + BC(\bar{A} + A)$$

$$= A\bar{B}C + BC \qquad [\because \bar{A} + A = 1]$$

$$= C(A\bar{B} + B)$$

$$= C(A + B) \qquad [\because A + \bar{A}B = A + B]$$

$$= CA + CB$$

$$= AC + BC$$